MIL-I-9443D
16 March 1972
SUPERSEDING
MIL-I-9443C(ASG)
26 April 1965

MILITARY SPECIFICATION

INDICATOR, TEMPERATURE, THERMOCOUPLE, ENGINE EXHAUST, TYPE MJ-2

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a single pointer thermocouple-type temperature indicator, designated type MJ-2.

1.2 Classification. Indicators shall be furnished in the following dial finishes, as specified (see 6.2):

Class 1 - Fluorescent dial
Class 2 - White dial

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

~~PPP-B-636~~	Box, Fiberboard

Military

MIL-P-116	Preservation, Methods Of
MIL-F-6836	Flange, Mounting Attachable, Aircraft Instrument
MIL-I-7071	Indicator, Temperature, Thermocouple, Hermetically Sealed, General Specification For

STANDARDS

Federal

FED-STD-595	Colors

FSC 6685

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Military

MIL-STD-129	Marking For Shipment And Storage
MIL-STD-130	Identification Marking Of US Military Property
MS33638	Case, Instrument, Flange Mounted, Aircraft
MS33639	Case, Instrument, Clamp-Mounted, Aircraft
MS35214	Screw, Machine-Pan Head, Cross-Recessed, Brass, Black Oxide, UNC-2A
MS35215	Screw-Machine-Pan Head, Cross-Recessed, Brass, Black Oxide, UNF 2A
MS35338	Washer, Lock-Spring, Helical, Regular (Medium) Series (IN./MM)

Air Force-Navy Aeronautical

AN961	Washer-Flat, Electrical

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules

(Applications for copies should be addressed to the Uniform Classification Committee, Room 202, Union Station, 516 West Jackson Blvd., Chicago IL. 60606.)

3. REQUIREMENTS

3.1 General. The requirements of MIL-I-7071 are applicable as requirements of this specification. Additional requirements shall be as specified herein.

3.2 Precedence. In the event the general specification and this specification conflict, this specification shall govern.

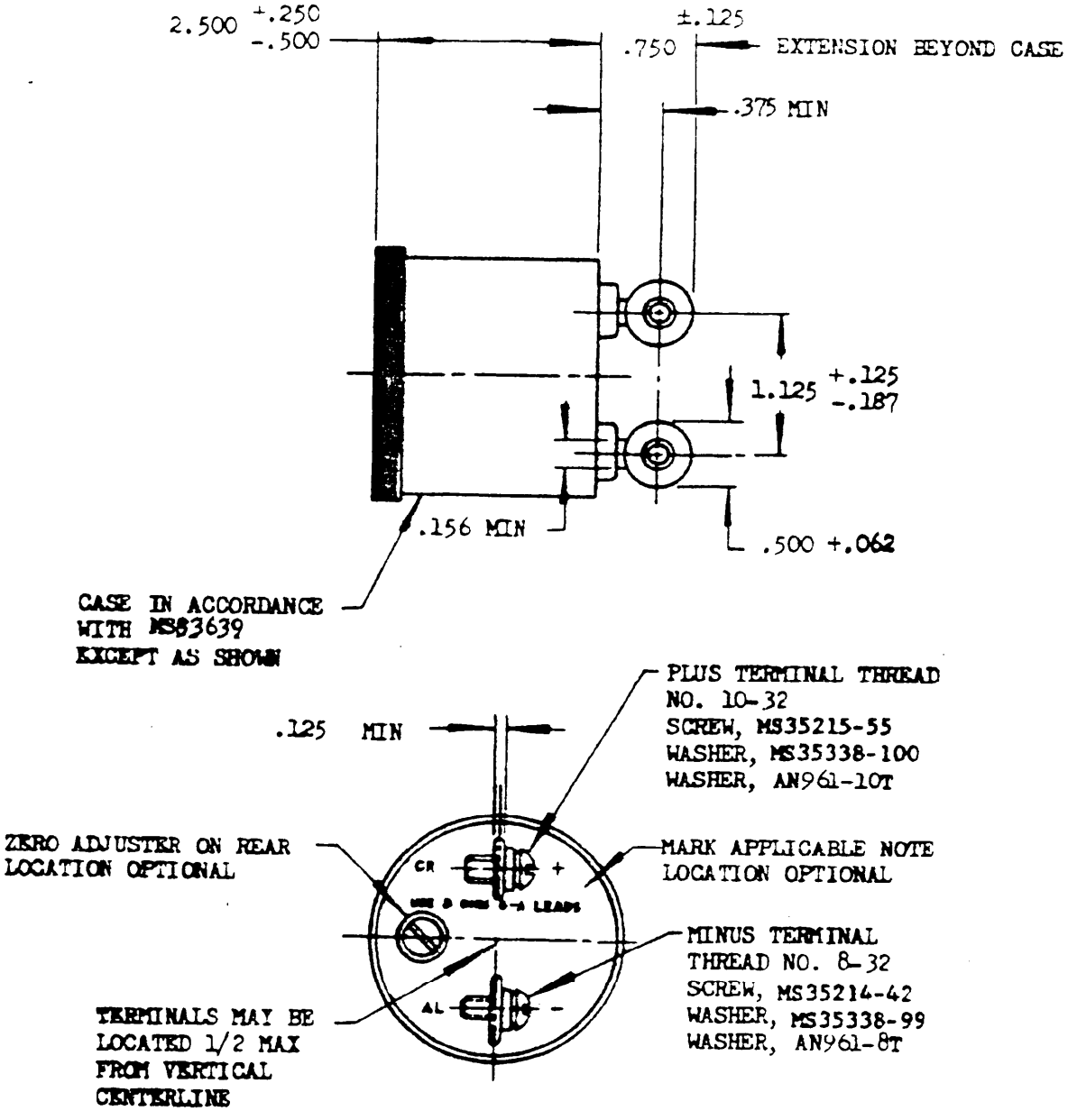
3.3 Qualification. The indicator furnished under this specification shall be a product which has been subjected to and which has passed the qualification tests specified, and which is listed on or approved for listing on the applicable Qualified Products List.

3.4 Design. The indicator shall conform to Figure 1.

3.5 Construction.

3.5.1 Case back. The back of the indicator case shall be distinctly marked to indicate "plus" for the chromel terminal stud and "minus" for the alumel terminal stud. The statement "Use 8-ohm C-A leads" shall be plainly marked on the rear of the case.

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BREAK SHARP EDGES .010 RADIUS.

DIMENSIONS IN INCHES.

FIGURE 1. Indicator details

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3.5.1.1 Case finish. The case shall be finished with lusterless black material, color No. 37038 of FED-STD-595. The finish shall be durable to withstand usage encountered in service.

3.5.2 Dial. The dial of the indicator shall conform to Figure 2.

3.5.3 Scale arc. The scale arc of the indicator shall be not less than 250 degrees. The mechanism shall not be geared to obtain the specified scale arc.

3.5.3.1 Scale distribution. The scale of the indicator shall be as linear as practicable and shall not be constricted or expanded by more than 1 angular degree between any 2 adjacent 20° Centigrade (C) scale graduations within the range of 200° to 900°C. The remainder of the scale shall not be constricted or expanded by more than 1.5 angular degrees between any 2 adjacent scale graduations. These scales are based on a minimum of 5 angular degrees nominal spacing between 20°C scale graduations.

3.5.4 Pointer. The width of the pointer 1/2 ±1/16 inch from the tip shall be 5/64 ±1/64 inch and shall taper to a point. The pointer shall be finished with fluorescent-luminescent material or white material, color No. 37875 of FED-STD-595, as specified by the procuring activity (see 6.2).

3.5.4.1 Pointer overswing. The pointer overswing shall be not less than 5 angular degrees deflection beyond each end of the scale.

3.6 Weight. The weight of the complete indicator shall not exceed 1.0 pound.

3.7 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

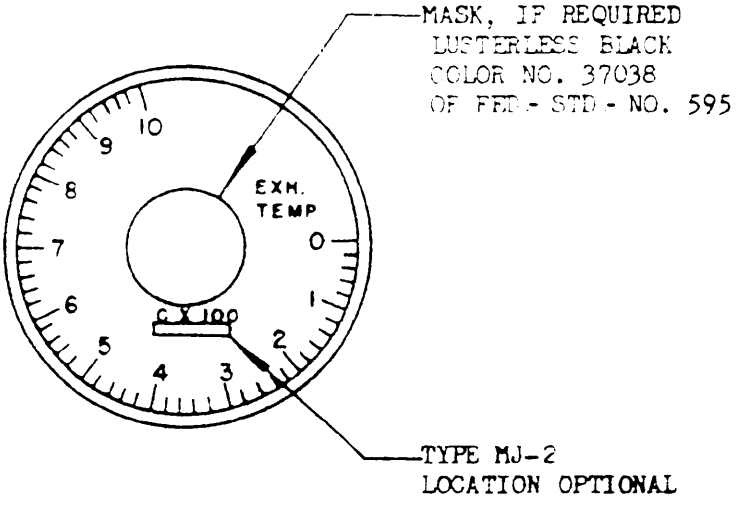
4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 General. Sampling, inspection, and testing of the indicator shall be as specified in MIL-I-7071 and as specified herein.

4.3 Test tolerances. The tolerances specified in Table I, not enclosed in parentheses, are applicable only when the indicators are subjected to all tests in the sequence specified herein. The tolerances specified in Table I, enclosed in parentheses, are applicable only when the indicators are subjected to a specific test, independent of sequence.

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MARKINGS	HEIGHT OR LENGTH ±0.005	WIDTH OF LINE OR GRADUATION ±0.005	MATERIAL OR FINISH
NUMBERS ALL MAJOR GRADUATIONS ALL MINOR GRADUATIONS LETTERING "EXH TEMP" "C X 100"	0.156 0.156 0.093 0.125 0.093	0.020 0.020 0.015 0.020 0.015	FLUORESCENT-LUMINESCENT MATERIAL OR LUSTERLESS WHITE, COLOR NO. 37875 OF FED.- STD - 595
BACKGROUND OF DIAL TYPE MJ-2	--- 0.062	--- 0.010	LUSTERLESS BLACK COLOR NO. 37038 OF FED.-STD.- 595

DIMENSIONS IN INCHES.

FIGURE 2. Dial and markings

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TABLE I. Exhaust Temperature Test Tolerances

Test point	Scale error at room temperature	Scale error at low temperature	Scale error at high temperature	Scale error at room temperature after vibration failure	Extreme low temperature exposure		
					Scale error at low temperature	Scale error at room temperature after low temperature	Scale error at room temperature after high temperature
°C	Tolerance ±°C	Tolerance ±°C	Tolerance ±°C	Tolerance ±°C	Tolerance ±°C	Tolerance ±°C	Tolerance ±°C
0	40 (40)	50 (50)	50 (50)	50 (50)	60 (50)	50 (40)	50 (40)
200	30 (30)	40 (40)	40 (40)	40 (40)	50 (40)	40 (35)	40 (35)
400	20 (20)	35 (35)	30 (30)	35 (30)	45 (40)	40 (30)	40 (30)
500	20 (20)	35 (35)	30 (30)	35 (30)	45 (40)	40 (30)	40 (30)
600	10 (10)	25 (25)	20 (20)	25 (20)	40 (30)	30 (20)	30 (20)
700	5 (5)	20 (20)	15 (15)	20 (15)	35 (25)	25 (15)	25 (15)
800	10 (10)	25 (25)	20 (20)	25 (20)	40 (30)	30 (20)	30 (20)
900	20 (20)	35 (35)	30 (30)	35 (30)	45 (40)	40 (35)	40 (35)
1,000	30 (30)	40 (40)	40 (40)	40 (35)	45 (40)	40 (35)	40 (35)

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4.4 Quality conformance tests.

4.4.1 Individual tests.

4.4.1.1 Scale error at room temperature. The scale errors at the test points specified in Table I shall not exceed the tolerances specified in Table I. When this test is conducted, the pointer shall be set to read 700°C when the millivoltage for 700°C, as specified in Table I of MIL-I-7071 for the ambient room temperature, is applied to the terminals of the indicator through the appropriate resistance in series with the indicator. If more accurate overall indicator accuracy can be obtained by setting the pointer to 700° ±5°C with the specified voltage for 700°C, the pointer shall be so positioned.

4.4.1.2 Position error. This test shall be conducted at any minor or major scale graduation between 200° and 900°C by applying test voltage. The maximum change in indication shall not exceed 25°C.

4.4.1.3 Friction. The maximum difference in reading, before and after tapping, shall not exceed 30°C.

4.4.1.4 Zero adjustment. Power shall be applied to produce a 700°C reading. When the indicator zero adjustment is operated throughout its range, the resulting pointer motion shall be not less than 40°C above and below the 700°C reading. The total motion of the pointer shall not exceed 200°C. The pointer shall be reset to the initial reading with the initial power applied.

4.4.1.5 Insulation leakage. The insulation resistance shall be a minimum of 20 megohms.

4.4.2 Sampling tests.

4.4.2.1 Scale error at high temperature. The scale errors shall not exceed the tolerances specified in Table I. If there is any tendency for the pointer to stick at the pointer stops, a light tap on the indicator shall be sufficient to free the pointer.

4.4.2.2 Scale error at high temperature. The scale errors shall not exceed the tolerances specified in Table I. If there is any tendency for the pointer to stick at the pointer stops, a light tap on the indicator shall be sufficient to free the pointer

4.4.2.3 Magnetic effect. When the indicator is subjected to the magnetic effect test, it shall not cause the magnetic compass to deflect more than 5 degrees.

4.4.2.4 Vibration error. The maximum amplitude of pointer oscillation throughout the frequency range shall not exceed a range of 25°C within the temperature range of 400° to 900°C and shall not exceed 30°C for the remainder of the scale. The maximum difference between the original position of the pointer before vibration and the position of the pointer during vibration shall not exceed 15°C between 400° and 900°C and shall not exceed 25°C for the remainder of the scale. Pointer oscillation shall not be considered as pointer variation. The diameter of the circular motion of vibration shall be 0.009 to 0.011 inch.

4.4.2.5 Vibration failure. The scale errors at room temperature, after vibration, shall not exceed the tolerance specificed in Table I. The zero adjuster shall not be reset after this vibration test.

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4.4.2.6 Resistance. The resistance of the indicator shall be 40 +5, -0 ohms.

4.4.3 Special tests.

4.4.3.1 Cold junction compensator lag. The indicator shall be connected to a chromel-alumel thermocouple (8 ohms total external circuit resistance), immersed in an ice bath at 0°C, and a millivoltage of constant value applied in series with the indicator and thermocouple to provide readings of 700°C. The indicator shall then be subjected to the conditions specified in MIL-I-7071 for increasing and decreasing temperatures. The indication of the indicator throughout the test shall at no time vary by more than 30°C from the indication obtained at room temperature.

4.4.3.2 Extreme low temperature exposure. The scale errors at low temperature shall not exceed the tolerances specified in Table I. The scale errors at room temperature, after low temperature exposure, shall not exceed the tolerances specified in Table I.

4.4.3.3 High temperature exposure. The scale errors at room temperature, at high temperature exposure, shall not exceed the tolerances specified in Table I.

4.5 Packaging, packing, and marking. Preparation for delivery shall be examined for conformance to Section 5.

5. PREPARATION FOR DELIVERY

* 5.1 Preservation and packaging. Preservation and packaging shall be Level A or C as specified (see 6.2c).

* 5.1.1 Level A. Each indicator shall be preserved and packaged in accordance with MIL-P-116, Method III, and placed in a snug fitting container conforming to PPP-B-636, Grade W6C.

* 5.1.2 Level C. Each indicator shall be preserved and packaged in a manner which will afford adequate protection against corrosion, deterioration and physical damage during shipment from supply source to the first receiving activity. The supplier may use his commercial practice providing it meets this requirement.

* 5.2 Packing. Packing shall be Level A, B or C as specified (see 6.2c).

* 5.2.1 Level A. Indicators preserved and packaged as specified in 5.1.1 shall be packed in weather-resistant shipping containers conforming to PPP-B-636. The grade of each shipping container, determined by the gross weight and size, shall be limited to the applicable special requirements table of the specification. Where practicable, the exterior shipping containers shall be of uniform shape and size and of minimum cube and tare. Closure and strapping shall be in accordance with the appendix of PPP-B-636.

* 5.2.2 Level B. Same as Level A except class domestic shipping containers may be used.

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* 5.2.3 Level C. The indicators shall be packed in a shipping container which will be acceptable to the carrier at lowest rates and insure safe transportation to the point of delivery. Containers shall comply with Uniform Freight Classification Rules and Regulations, or other regulations of other carriers as applicable to the mode of transportation.

* 5.3 Marking. In addition to any other markings required by order or contract, the interior and exterior containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The type MJ-2 indicator is intended for use in indicating exhaust temperatures on jet engines.

6.1.1 The temperature indicator can be used on instrument panels requiring case mounting in accordance with MS33638, by the application of an instrument mounting bezel in accordance with MIL-F-6830.

* 6.2 Ordering data. Procurement documents should specify:

a. Title, number, and date of this specification.

b. Dial and pointer finish (see 1.2 and 3.5.4).

c. Levels of packaging and packing required (see Section 5).

* 6.3 With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the San Antonio Air Materiel Area, Service Engineering Division, ATTN: MMEE, Kelly AFB, Texas 78241, and information pertaining to qualification or products may be obtained from that activity.

6.4 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Air Force - 82

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Preparing Activity:
Air Force - 82

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A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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